Car No. : 008

Design Report for SAE BAJA 2021

Team Grease Monkeys

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Abstract

SAE BAJA, the intercollegiate ATV design competition provides a platform for undergraduate students to apply their principles of engineering science to exhibit their proficiency in the automotive world. The objective of this competition is to design and fabricate an All Terrain Vehicle (ATV) that can withstand rough terrains and is also viable to be manufactured for consumer sale. The criteria considered in the design are safety, performance, manufacturability, cost and ergonomics.This design report contains design objectives, vehicle concepts, design approach, important features and application of analysis. The vehicle was designed, modelled and assembled using SolidWorks 2021 to assist in finding potential assembly problems. Finite Element Analysis was performed using ANSYS 2021R2 for roll cage and other components. It reflects the combined efforts of all the team members.

Introduction

The major change in the design was switching from a RWD to a 4WD design. The main objective was incorporation of 4WD components while keeping in mind the safety parameters. A new chassis model was made and special focus was given to the rulebook while incorporating 4WD components. Modelling was done in CAD software SolidWorks 2021 to obtain a model with maximum geometric details. Finite Element Analysis was done using ANSYS 2021R2 software. Based on the analysis results, the design was remodelled and retested and a final design was adopted. Dynamic analysis was done in Lotus suspension analysis software. Theoretical calculations of performance characteristics were also done for each system and components. Extensive weight reduction techniques were followed at every stage of the design to improve performance without sacrificing structural integrity. During the entire design process, consumer interest through innovative, inexpensive, and effective methods was always the primary goal.

Overall Design Objectives:

1. Reduce unsprung mass and total mass
2. Better manufacturability
3. Ergonomics and Safety of driver
4. Implementing 4WD system
5. Quality components at reasonable cost
6. Market Availability of OEMs
7. Having maximum clearances among components

This year the main objective was to incorporate 4WD into our vehicle ensuring all parts are running optimally. The components which were added were transfer case, propeller shaft, differential and front axles. Changes had to be made to accommodate all of these components. Suppose the shocks on the front wheel had to be shifted to the upper A-Arm instead of the lower A-Arm to make space for the axle.

Rollcage

The design objectives were to ensure safety, optimize weight, improve ergonomics and meet the requirements put forward by other subsystems, especially the changes due to 4WD under the constraints provided by the rulebook. It was also assumed that the frame would be manufactured on campus with the tools available to us and the logistical constraints in acquiring the material needed.

Material Selection

AISI 4130 chromoly tubing (yield strength as provided by the vendor= 650 MPa) was chosen as the material to reduce weight without compromising strength.
The following dimensions were chosen:

Table 1 Tube dimensions and bending stiffness/strength

OD	ID	Bending Stiffness	Bending Strength
0.0254	0.0234	1170.824	195.624
0.03175	0.02845	3631.449	485.402

Design began with review of the previous year's design and planning changes in the current year design- 4WD accommodation, ergonomics and driver safety adhering to the rulebook. We decided to continue with the no nose design for better ergonomics. The overall height of the roll cage and the height of the SIM was increased to accommodate the intermediate shaft through the cockpit and ensure driver safety. The width and length of the nose were increased to house the front differential. SIM was equipped with a 2-D double bend to prevent violation of knee rule, its distance of the bend from RRH was decided from the CAE analysis to maximize the FOS. This bend was kept 2D to reduce errors during the manufacturing. Geometry of front FAB and SIM-LFS bracing members was modified on the basis of CAE. HDPE will be used for Belly Pan instead of galvanized iron to reduce weight.

Roll Cage Analysis

CAE based design process followed for the Rollcage-

1. Chassis geometry modeled on Solidworks.
2. Static analysis of Chassis on ANSYS 2020 R2
3. Geometry modified iteratively where FOS was low.
4. Chassis geometry modified as per the requirement of transmission/suspension subsystems.
5. Final static analysis of chassis based on weight obtained from other subsystems.

Assumptions:

1. The material is orthographic isotropic.
2. Mass of vehicle + driver = M = 250 kg
3. Velocity of vehicle (V)= 45 kmph or 12.5 m/s
4. Impact time = 0.15s (Tr) for rigid bodies and 0.3s (Td) for deformable bodies
5. Impact force = MV/T; T = Tr or Td

Quadrilateral Surface mesh was chosen to mesh the roll cage since the length to thickness ratio is high. The mesh was refined near the joints.

Table 2 Mesh Metrics

Element Size	No. of Nodes	No. of Elements	Element Shape	Aspect Ratio	Jacobian Ratio
5.00 mm	260251	260807	Quadrilateral	1.0607	0.96951

Roll Over

The car was assumed to fall from a height of 8 feet, while rolling, and land upside-down with the RHO and LC's hitting the ground first. The ground was assumed rigid; thus, time of collision was 0.15sec. All suspension points were assumed rigid in translation and rotation about X-axis and Z-axis only, were permitted.

S = 8 ft = 2.4384 m

$v^2 = 2as$; From the relation, v = 6.916 m/s

Torsional Analysis

4g vertical force was applied on the front-left and rear-right shock attachment point. Front-right and rear-left suspension points were kept fixed.

Torque = M*4g*distance between points

Angular Displacement = deformation/ distance

Torsional Rigidity = Torque/ Angular Displacement

Results and Calculations:

Distance between points = 1394 mm

Angular Displacement = 0.0039 rad = 0.223 deg

Torque = 250*4g*1.394 = 13.675 kNm

Torsional Rigidity of the chassis = 61.321 kN/degree

Table 3 Forces applied and CAE results

Impact Scenario	Impact Time (s)	Impact Force (kN)	Max Stress (Mpa)	Max Deformation (mm)	Min FOS
Front Impact	0.15	20.8	282.7	0.884	1.627
Rear Impact	0.15	20.8	220.9	0.523	2.083
Side Impact	0.3	10.4	204.5	0.57	1.797
Roll Over	0.15	11.5	218.9	1.894	2.102
Bump (4g force on one suspension mount)	-	9.8	374.6	3.33	1.228
Torsional	-	9.8	335.6	5.502	1.371

Suspension

Objectives and research

Design considerations

1. To design a suspension geometry which is an appropriate trade-off between riding and handling characteristics
2. To minimize toe variation, camber variation and tyre scrub during wheel travel.
3. To achieve anti squat characteristics

Design Process

The team started the design process by revisiting previous years designs, after which basic car parameters were set. Type of suspension for F and R were fixed. A rough model was built into Vsusp software which was followed by making 2D drawings of front and side view geometry in solidworks. Parameters like static camber, Kpi, caster were set. Shock calculations were done in matlab and the model was refined. The rough points obtained from Solidworks were imported into ansys and variation of camber, toe,caster , Kpi was analysed for bump and droop.(cornering,roll). Necessary geometry changes were made. After fixing the final points, wheel assembly components like upright, hub were designed and interferences were checked. These were imported into ansys and analysed for max stress developed, Fos, deformation at various scenarios. The components were optimised and final assembly was made.

Front suspension

Double Wishbone was chosen as the front suspension because of its simplicity and high controllability over various parameters to achieve less variation in toe and camber angles over the wheel travel. The shocks are connected to the upper wishbone due to the possibility of interference with the drive shaft. The suspension geometry was finalized on lotus suspension. Analysis and subsequent simulations were carried on ADAMS Car's suspension module for kinematic and dynamic analysis respectively. Primary rods of Chromoly 4130 were used to manufacture the control arms. Pin Joints have been used to connect both the upper and lower control arms to the chassis. Bushes were used to connect the pivot joints to the M10 bolts for the control arms. Fox Float 3 progressive air shocks are used for both the front and rear suspension system. The motion ratio for the front was kept to be 0.7 to achieve a wheel travel of 5.1 inches.

Rear suspension

The natural frequency for the rear suspension was taken to be 20% higher than the front suspension. Based on multiple parameters such as the flexibility of roll centre height, need for camber change, packaging constraints and modifications for 4wd, the considered suspension systems in the rear were, double A wishbone, H-arm, trailing arm and semi trailing arm, out of which semi trailing arm was considered to be the best choice. Static, Dynamic and Multi body analysis was done in Ansys and ADAMS software, based on which the design was further optimised for an appreciable FOS and scope of weight reduction. To avoid the phenomenon of Rod ends in bending, encapsulated spherical bearings were used for the ends of the Semi Trailing arm.

Wheel Assembly

Design Objectives

1.Reduce unsprung mass by using materials with high strength to mass ratio.

2.Compact packaging for small scrub radius.

3.High misalignment angle spacers with rod ends for high wheel travel .

4.Assembly such that it's easy to access caliper and rotor for maintenance.

5. Avoiding interferences between 4WD transmission and suspension components.

Forces

1. The car is assumed to undergo free fall of 2m and lands on the front two tires which gives a

net force of 8g and the maximum bump force on one wheel was 4g.
2. Taking force balance on a banked road with inclination 45 degree for maximum force and coefficient of friction to be 0.6,cornering force was 2775N.
3. Braking torque of 150 Nm was applied by taking four radial forces on the hub 800 N.
4. Driving torque was applied on rear hub by taking four radial forces of 1600N on the rim hole.
5. The maximum force developed in the shock was calculated to be 4443N based on maximum travel of 4.5 inches . This force was used in case of bump for arms.

Analysis

1. Von mises criteria was used for calculating stresses
2. Multiple S-N curves were used for fatigue instead of using fatigue correction criteria.
3. For uprights and hubs tetrahedron mesh was used since they are complex geometries and have varying stresses over short distances.
4.For A arms and semi trailing arms surface mesh was used since the length to thickness ratio is high.

Steering

Design Objectives

1. To provide directional stability of the vehicle when going straight ahead.
2. To facilitate straight ahead recovery after completing a turn.
3. To reduce the steering torque and achieve a good Ackermann percentage. .
4. To reduce turning radius.
5. To minimize bump and roll steer.
6. To minimize toe variation, camber variation and tyre scrub during wheel travel.
7. To design a suspension geometry which is an appropriate trade-off between riding and handling characteristics.

Calculations

Ackerman Angle (α)

Tan α= wheelbase/[(wheelbase/tan(angle of outside lock)-wheel track]
α=45.50 degrees

Turning radius

$$\cot\delta = (\cot\delta_o + \cot\delta_i)\frac{1}{2}$$

$$R = \sqrt{a_2^2 + l^2\cot\delta}$$

Where a2 is distance b/w cg and rear wheel centre,l is wheelbase,delta is average angle of turn
=2.256 m

Steering Ratio = 4.76

Steering Torque

Torque due to cornering force(torque due to caster)(N mm)= 25780 N mm

traction torque(N mm)= Normal force on tyre N (worst case) x coefficient of friction x scrub radius= 550 x0.7 x27.55= 10606.75
Total Torque= 36387 N mm.

Steering Rack Position

The rack can have two positions. It can either be in front of the front wheel center line or behind it. If the rack is placed forward of the front axle line it can be mounted easily on the frame giving a wide range for choice of heights. If the rear steer is chosen, the drive shaft and its casing does not provide enough space for the steering rack . Therefore ,considering the 4 wheel drive model, fixing the rack in front of the axle line is better from both a geometrical and packaging viewpoint. Hence it is decided to have the rack positioned in front of the front axle line.

Design Approach

1 Setting basic car parameters (TW and WB).
2 Selection of steering geometry and steering mechanism.
3 Coordinating with suspension to get tie rod length and steering arm hard points.
4 Assuming some steering parameters based on existing data.
5 Perform matlab calculations of ackermann steering model.(kinematic).
6 Simulating geometry in solidworks.
7 Checking interferences with various components.
8 Keeping appropriate clearances for differential.
9 Try to achieve perfect Ackerman geometry.
10 Simulate in lotus and try to minimise roll and bump steer.
11 Modelling steering components.
12 Analysis and optimization.

Drivetrain

Design Objectives

1. Implementation of full-time 4WD design.
2. Increase gradeability performance.
3. Enhance acceleration performance.

A full time 4WD Drivetrain system has been designed. The output shaft from the engine is directly coupled to the CVT via keyway shaft which is connected to a self-centring secondary clutch through a rubber belt. The driven clutch is attached to a transaxle gearbox consisting of spur gears and locked differential. The intermediate shaft of the gearbox is coupled with the transfer case consisting of a 90 degree power transferring bevel set functioning as a locked differential, which is connected to a propeller shaft that transmits power to the front axle through an open differential. Inboard tripod joints and outboard Rzeppa joints have been used in the front and rear half-shafts.

Gearbox And Transfer Case

A new single speed two-step reduction gearbox was designed and custom-made. To calculate the gear ratio, the following approach was adopted:

Design Approach And Calculations

A theoretical top speed of 60 km/h was aimed for.

$$v_{max} = (d_w/2) * w_w$$

$$w_w = w_E * CVT_{overdrive} * GR * \eta$$

Where, Wheel diameter(d_w)= 23 in (0.5842 m)
Engine angular velocity(w_E)= 3800 rev/min
Drivetrain efficiency(η)= 0.81
CVT overdrive ratio= 0.7

The gear ratio(GR) came out to be 8.07.

However, considering gradeability performance and equal angular velocity(w) requirements at the front and rear axle due to a locked transfer case, an equal torque split was supposed to be given between front and rear axles and number of teeth in the respective gears to be decided accordingly.
Minimum number of teeth allowed in the pinion gears was calculated using the formula:

$$N_P = \frac{2k}{(1+2m)\sin^2\phi}\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)$$

Hence, with a pressure angle of 20° and module of 2.5 for the spur gears, the gear ratio was finalised as:
Step1: 54/18= 3
Step2: 56/20= 2.8
Final gear ratio= 3*2.8= 8.4

To have an equal torque split at the front axle, gear ratios of 1.4 and 2 were fixed at the transfer case bevel gear set and the front open differential respectively. Hence, the transmission ratio (excluding CVT ratio) at front and rear axle is 8.4. Overall, compared to the previous year, the gearbox design has been optimized (from 11.81 to 8.4) and weight has been reduced.

Half shafts:

OEM solid half shafts made of EN36 (carbon steel) have been used. The diameter was finalised using the following formula:

$$T = \pi * \tau * (d^3/16)$$
$$T = F * r_w/2$$
$$\tau = Ultimate\ Tensile\ Strength/2$$

Ultimate tensile strength of EN36 is 1230 N/mm^2. Total load (F) exerted on the wheels was estimated to be 2206.06 N. The minimum diameter of the half shaft came out to be approximately 14mm. Hence, taking the factor of safety as 1.57, half shafts of diameter 22mm have been selected.

Analysis and choice of material:

A detailed analysis of all the drivetrain components has been done and material selection and optimization has been done accordingly. Total deformation, equivalent stress and factor of safety have been evaluated using ANSYS 2021R2 software for all the components. Face width of the gears was determined using AGMA 2001-D4, after considering various factors like overload factor, Dynamic factor, Size factor and Geometry factor. Solid shafts with steps were considered for initial design for proper alignment of the components on the shafts. Based on the loads transmitted, bending moment diagrams were plotted for each shaft to determine its diameter at each point. When considering the shaft length, an approximate length for the bearing has been considered on the shaft. The design of all the components is within safety limits and the material EN36 is an appropriate choice.

Drivetrain Specifications Comparison:

Table 4 Drivetrain Comparison of last year and new design

Specification	BAJA 2021	BAJA 2022
Top speed	53.14 km/h	54.61 km/h
Maximum acceleration	5.34 m/s^2	6.02 m/s^2
Gradeability	67% (33.8°)	75% (36.9°)
Gearbox gear ratio	11.81:1	8.4:1
CVT gear ratio	0.43:1 to 3:1	0.7:1 to 3.9:1

Braking

Design Objectives

1. To lock all four wheels at the same time in static conditions as well as at-speed on paved and unpaved surface
2. To improve ergonomics wrt pedal assembly and brake lines as compared to last year's braking system.
3. To reduce weight using proper optimization techniques
4. To improve performance by using an outboard braking system over inboard from previous years

Design Approach

The brake Rotors and pedal ratio was fixed first. Once these were decided, calculations for required braking force and torque could be done, and hence calipers chosen; in such a way that generated torque and force is sufficiently higher than required. Optimisation of pedal and rotor was done at the end.
Braking values:
Weight distribution ratio=35:65
Effective rotor dia =217 mm
Caliper bore dia front =32mm
Caliper bore dia rear =35.5mm
Pedal ratio =5:1
Master cylinder bore dia : 0.647 mm
Force on master cylinder= 2000N
Braking circuit : X split

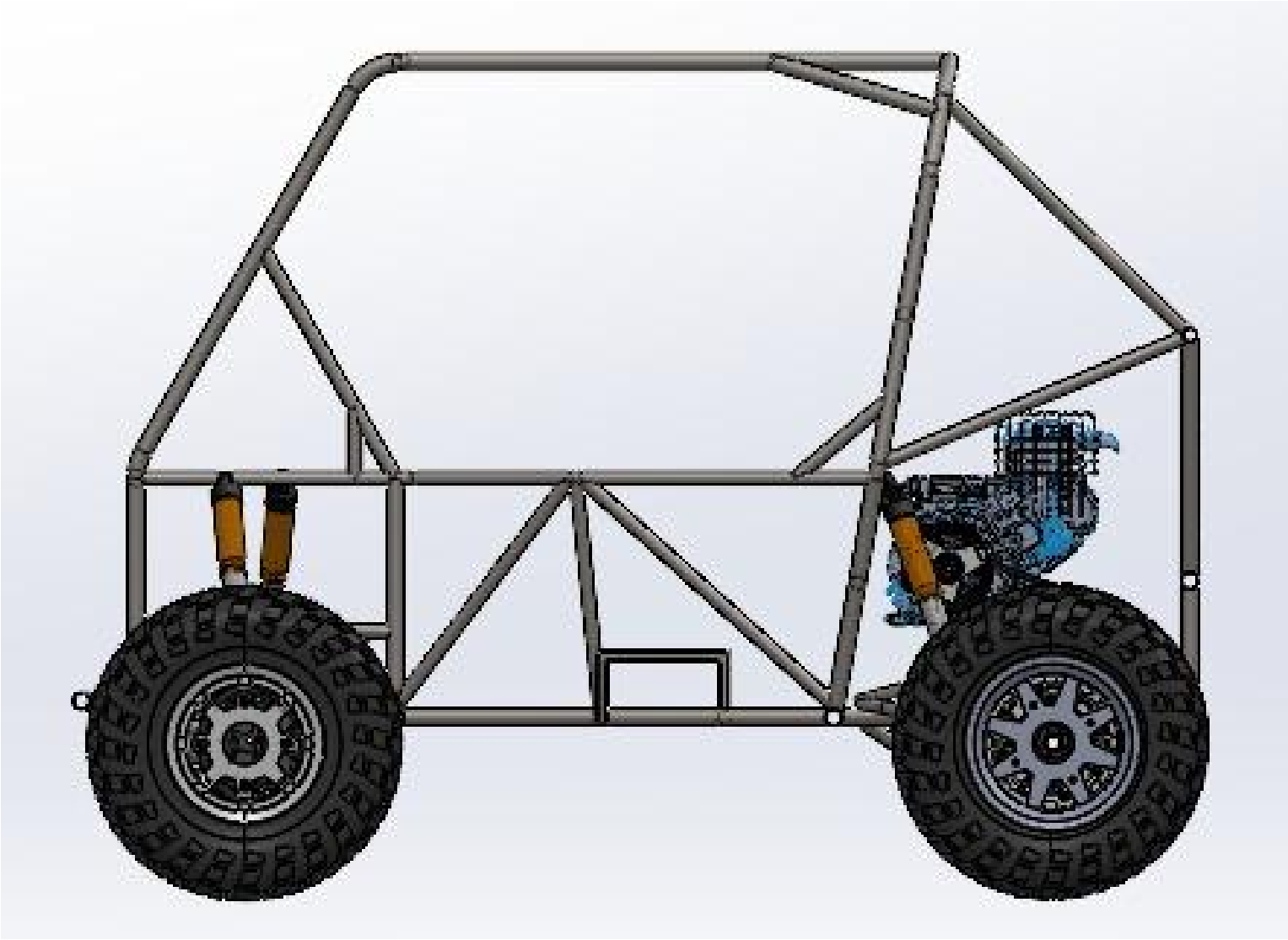
Table5. Braking values:

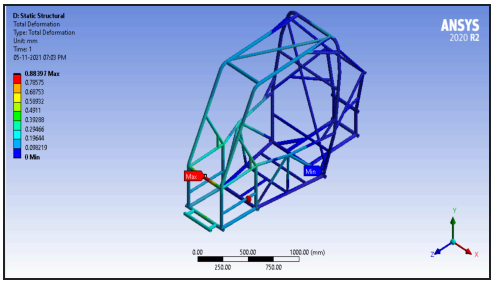
dynamic weight transfer	79.941 kg
weight front wheels kg	167.441 kg
weight rear wheels kg	82.558 kg
front reaction force N	1674.413 N
rear reaction force N	825.587 N
front axle braking force	1172.089N
rear axle braking force	577.911 N
torque req per wheel front	**171.184 Nm**
torque req per wheel rear	**84.404 Nm**
brake force on rotor front	4445.051
brake force on rotor rear	3611.769
torque gen per wheel front	**436.948**
torque gen per wheel rear	**355.0369**

Calipers are chosen such that torque generated is sufficiently more than torque required.

References:

1. SAE mBAJA Rulebook 2022
2. Shigley's Mechanical Engineering Design (McGraw-Hill Series in Mechanical Engineering) Tenth Edition.
3. Gillespie, T. D. (1992). *Fundamentals of vehicle dynamics*. SAE International.
4. Race Car Vehicle Dynamics (William F. Milliken, Douglas L. Milliken)
5. Tarkeshwar Patil, Rajeshwari Patil, 2016, Design and Analysis BAJA ATV's Half Axle, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 05, Issue 11 (November 2016), http://dx.doi.org/10.17577/IJERTV5IS110074

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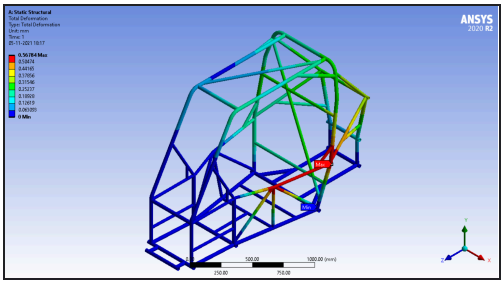
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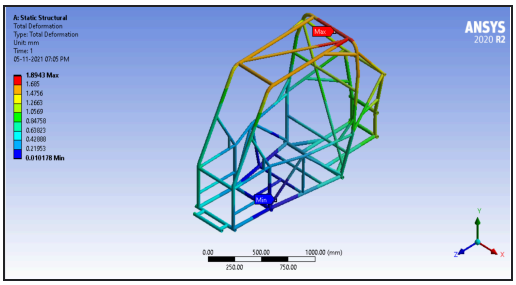
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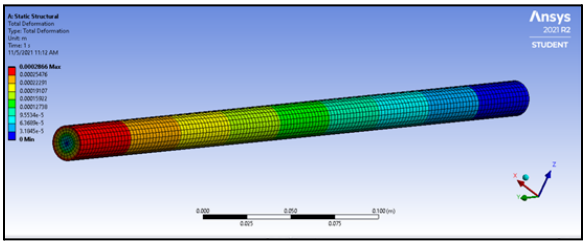
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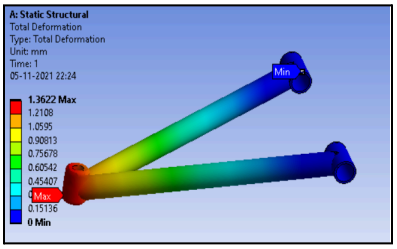
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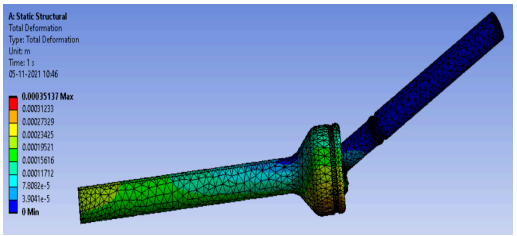
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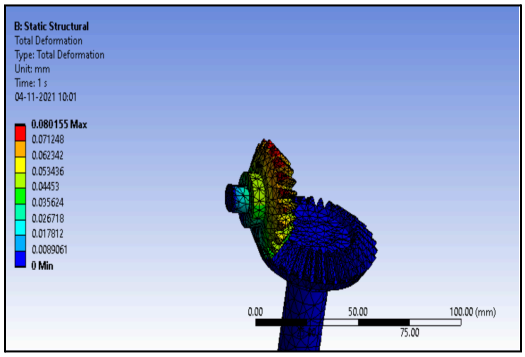
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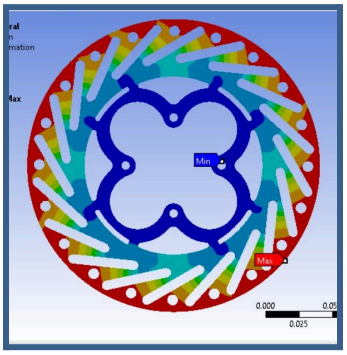
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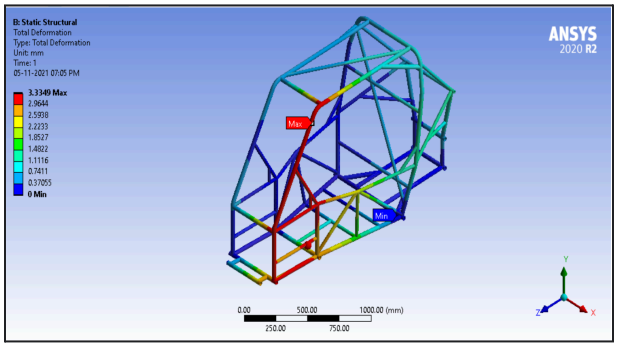
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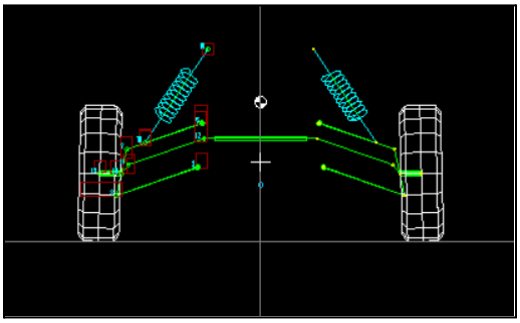

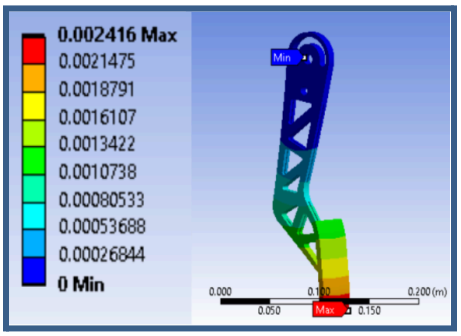
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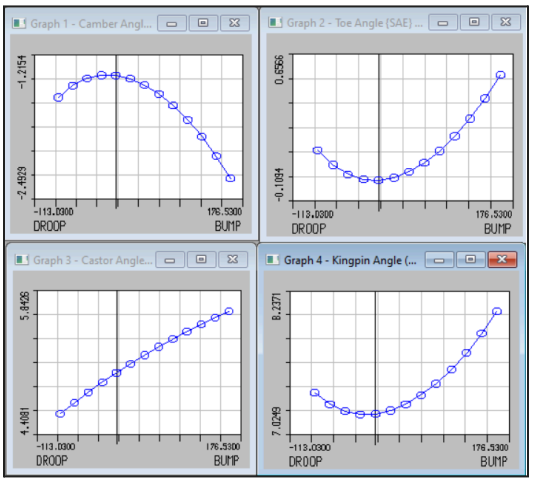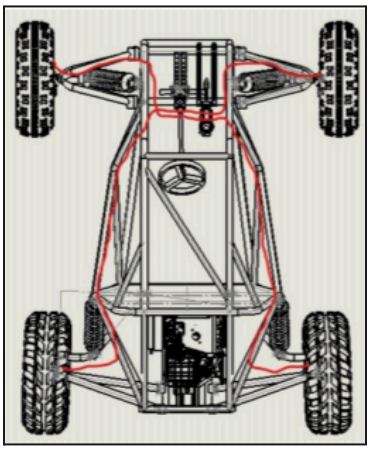
Graph 1 - Camber Angl...
-1.2154
-2.4923
-113.0300
176.5300
DROOP
BUMP
Graph 2 - Toe Angle (SAE) ...
0.5566
-0.1084
-113.0300
176.5300
DROOP
BUMP
Graph 3 - Castor Angle...
5.8426
4.4081
-113.0300
176.5300
DROOP
BUMP
Graph 4 - Kingpin Angle (...
8.2271
7.0249
-113.0300
176.5300
DROOP
BUMP